

Business Calculus (MATH 2350)
Summer 2018
Thursday, 05/31/2018
Quiz 3: Section 11.2,11.3

Name (Print): _____

Score: _____ / 100

Section 11.2

1. 15 points Let $f(x) = 9e^x + x^5$. Find the derivative $f'(x)$.

Using the linearity of the derivative, the power rule, and the fact that $\frac{d}{dx}e^x = e^x$, we yield

$$f'(x) = 9e^x + 5x^4.$$

2. 20 points Let $f(x) = \ln(x^6) + 3x^3 + 80$. Find the derivative $f'(x)$.

Hint: $\log_b(x^n) = n\log_b(x)$.

First, we re-write $f(x)$ by using the property of logarithms given in the hint. This gives

$$f(x) = 6\ln(x) + 3x^3 + 80.$$

Now, using the linearity of the derivative, the power rule, and the fact that $\frac{d}{dx}\ln(x) = \frac{1}{x}$, we yield

$$f'(x) = \frac{6}{x} + 9x^2.$$

3. 20 points Let $f(x) = 3x + \ln(3x) + e^{3+x}$. Find the derivative $f'(x)$.

Hint: $e^{r+t} = e^r e^t$ and $\log_b(rt) = \log_b(r) + \log_b(t)$.

First, we re-write $f(x)$ by using the properties of logarithms and exponents given in the hint. This gives

$$f(x) = 3x + \ln(3) + \ln(x) + e^3 e^x.$$

Now, using the linearity of the derivative, the power rule, and the fact that $\frac{d}{dx} \ln(x) = \frac{1}{x}$ and $\frac{d}{dx} e^x = e^x$, we yield

$$f'(x) = 3 + \frac{1}{x} + e^3 e^x = 3 + \frac{1}{x} + e^{3+x}.$$

Section 11.3

1. 15 points Let $f(x) = 3x^4e^x$ Find the derivative $f'(x)$.

Let $g(x) = 3x^4$ and let $h(x) = e^x$. Then, $f(x) = g(x)h(x)$, and we may apply the product rule to obtain

$$f'(x) = g'(x)h(x) + g(x)h'(x).$$

Calculating, we obtain that $g'(x) = 12x^3$ and $h'(x) = e^x$. Thus, we yield that

$$f'(x) = 12x^3e^x + 3x^4e^x.$$

2. 15 points Let $f(x) = \frac{1 + e^x}{1 - x}$ Find the derivative $f'(x)$.

Let $g(x) = 1 + e^x$ and let $h(x) = 1 - x$. Then, $f(x) = \frac{g(x)}{h(x)}$, and we may apply the quotient rule to obtain

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{(h(x))^2}.$$

Calculating, we obtain that $g'(x) = e^x$ and $h'(x) = -1$. Thus, we yield that

$$f'(x) = \frac{e^x(1 - x) - (-1)(1 + e^x)}{(1 - x)^2}$$

If we wish to simplify this (which you are not required to do for full credit), we obtain

$$f'(x) = \frac{e^x(1 - x) + (1 + e^x)}{(1 - x)^2} = \frac{e^x - xe^x + 1}{(1 - x)^2}.$$

3. 15 points Let $f(x) = \frac{x^2 + 2}{2x^2 + x}$ Find the derivative $f'(x)$.

Let $g(x) = x^2 + 2$ and let $h(x) = 2x^2 + x$. Then, $f(x) = \frac{g(x)}{h(x)}$, and we may apply the quotient rule to obtain

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{(h(x))^2}.$$

Calculating, we obtain that $g'(x) = 2x$ and $h'(x) = 4x + 1$. Thus, we yield that

$$f'(x) = \frac{(2x)(2x^2 + x) - (x^2 + 2)(4x + 1)}{(2x^2 + x)^2}$$

If we wish to simplify this (which you are not required to do for full credit), we obtain

$$f'(x) = \frac{x^2 - 8x - 2}{(2x^2 + x)^2}.$$

4. 10 points (bonus) Let $f(x) = \frac{(x^3 + 9x^2 + 4)(x^4 + x^3 + x + 1)}{(x^2 + 1)}$. Find the derivative $f'(x)$.

Let $g(x) = x^3 + 9x^2 + 4$ and let $h(x) = x^4 + x^3 + x + 1$. Then, $f(x) = \frac{g(x)}{h(x)}$, and we may apply the quotient rule to obtain

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{(h(x))^2}.$$

To compute $f'(x)$, we require the derivatives of $g(x)$ and $h(x)$. We can see that $h'(x) = 4x^3 + 3x^2 + 1$ by using the power rule, but $g'(x)$ would either require a lengthy expansion or the use of the product rule, which is much more desirable.

Let $q(x) = x^3 + 9x^2 + 4$ and $l(x) = x^4 + x^3 + x + 1$. Then, $g(x) = q(x)l(x)$, and we may use the chain rule to obtain $g'(x) = q'(x)l(x) + q(x)l'(x)$. Calculating, $q'(x) = 3x^2 + 18x$ and $l'(x) = 4x^3 + 3x^2 + 1$. Thus, we yield that

$$g'(x) = (3x^2 + 18x)(x^4 + x^3 + x + 1) + (x^3 + 9x^2 + 4)(4x^3 + 3x^2 + 1).$$

This then allows us to reach our final answer, which is given by

$$f'(x) = \frac{[(3x^2 + 18x)(x^4 + x^3 + x + 1) + (x^3 + 9x^2 + 4)(4x^3 + 3x^2 + 1)](x^2 + 1)}{(x^2 + 1)^2} - \frac{(x^3 + 9x^2 + 4)(x^4 + x^3 + x + 1)(2x)}{(x^2 + 1)^2}$$

If we wish to simplify this (which you are not required to do for full credit), we obtain

$$f'(x) = \frac{5x^8 + 40x^7 + 34x^6 + 70x^5 + 59x^4 + 20x^3 + 38x^2 + 10x + 4}{(x^2 + 1)^2}$$